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Does Superficial Attraction Hinders Sustainable Relationships with Digital Humans? Understanding Customer Commitment on Digital Humans in e-Services from the Utilitarian and Hedonic Perspectives

Abstract: The rapid development of artificial intelligence technology has made digital humans (DHs) important interactive subjects in e-service systems. However, recent studies have reported that customers rarely accept DHs in e-service as people used to think. Given that DHs function as sociotechnical systems (STS), considering relationship management between DHs and customers in e-services to address these issues is necessary. To address the above concerns, this study uses the utilitarian and hedonic framework to investigate the construction mechanism of customer commitment (on DHs) in e-services. After three waves of data collection, a total of 703 questionnaires were collected. Multilevel modeling (MEM) and the response surface method (RSM) were used to analyze the data. The results demonstrate that customer commitment is constructed by a mixture of utilitarian and hedonic factors, in which customer experience plays a mediating role in e-service. However, DHs failed to construct continuance commitment in this study. This study not only provides corresponding theoretical contributions to the study of DHs, customer commitments, and human–machine relationships (HRMs) in e-services but also provides practical insights into the construction of commercial scenarios such as AI customer service and intelligent interaction systems.

Keywords: Digital human (DH); Customer commitment; Sociotechnical system; Utilitarian and hedonic framework; Customer experience

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Introduction

Numerous studies have indicated that digital human have attempted to embed themselves into the e-service loop by reshaping the interaction between customers and technology both utilitarianly (Wonggom et al., 2020) and hedonically (Butt et al., 2021). However, studies have indicated that customers rarely accept the fact that DHs have attempted to embed in e-services. Several scholars have already questioned DHs' endorsement in advertisements (Xue et al., 2025) and interaction in livestreaming (Xiao et al., 2025). Uncanny valley theory reveals that people's satisfaction is nonlinear. Specifically, people may be fearful or uncomfortable with DHs under certain circumstances (Liu & Wang, 2025).

To address this issue, analyzing the role and function of DHs in e-services is necessary. On the one hand, DHs function as "social actors" to participate in the construction of customer experience in e-service (Casaló et al., 2020; Mehmood et al., 2024). To some extent, a DH is a certain sociotechnical system (STS) in e-service (Ashish Singla et al., 2023; Kang et al., 2024). On the other hand, the function of DH in e-service is complex and hybrid with utilitarian and hedonic aspects (Bae et al., 2024).

On the basis of these two facets of DH in e-service, improving improve the customer's online experience is necessary (Ainhoa Apraiz et al., 2023; Huang & Rust, 2021). Prior studies have suggested that customer relationship management (CRM) lies at the center of such improvement (Nguyen et al., 2022; Ozuem et al., 2024; Youn & Jin, 2021). On the basis of these points, the rise of DH in e-services requires us to rethink CRM between DHs and customers.

However, current studies fail to reveal how such a relationship differs from conventional relationships and what roles and effects DH and customers experience in CRM (Bae et al., 2024; Seymour et al., 2024; Zhu & Yi, 2024). Moreover, given that commitment is crucial for CRM in e-services on the basis of commitment–trust theory (Badrinarayanan & Ramachandran, 2024; Patrucco et al., 2024) and shaped by economic and emotional aspects (Butt et al., 2021; Chunlin Yuan et al., 2022), the scarcity of discussion on customer commitment (on DHs) construction and management highlights the specific research gap of this study.

Aligned with such analysis, STS offers appropriate theoretical insight for solving this issue. STS incorporates human–technology interactions and an entire organization system formed by such interactions (Pasmore et al., 2019). In addition, the use of a utilitarian and hedonic framework facilitates addressing this issue.

On the basis of these viewpoints, this study uses the utility and hedonic framework to explore the relationship between DHs and customers. In this paper, a questionnaire survey method is adopted, 703 questionnaires are collected, and multilevel modeling (MEM) and the response surface method (RSM) are used to verify the theoretical model. The research

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conclusion shows that different characteristics of DHs have different effects on the establishment of customer commitment when different stimuli are generated for customers. The findings of this study facilitate improvements in the comprehension of DHs both theoretically and practically.

Theoretical background and hypothesis development

DHs in e-service: STS and customer acceptance

DH generally refers to a type of artificial virtual human that is artificial, has a real-life image, and is driven by AI or AI agents (Felnhofer et al., 2023; Lyytinen et al., 2021). In other words, DHs in their current form can be seen as the personification of AI, and they have a human appearance image that can interact with customers in real time. Following this viewpoint, previous studies indicate that unlike existing avatars (Thomas et al., 2025), DHs emphasize the concept of humanity and possess specific social roles (Berente et al., 2021; Raisch & Krakowski, 2021). For instance, Lyytinen et al. (2021) specifically suggest that DHs not only have anthropomorphic forms but also represent an entirely new STS. STS refers to organizational work systems that include both human and technological components. Within this system, humans and technology interact and influence each other, driving the transformation and development of organizational work systems (Bostrom et al., 2009). Therefore, as a type of STS, DHs incorporate both human and technological elements. Only through the integration of humans and technology can DHs fulfill their social functions.

This strong socio-function highlights the business value of DH, especially in e-service. E-service refers to the delivery of services by new media such as the internet or Metaverse, ranging from selling goods with services to providing pure services (Blut, 2016; Rowley, 2006). Specifically, website design, customer fulfillment and security consisted of this e-service quality (Blut, 2016). Considering metaverse advancement, DHs began to embed an e-service loop by exerting their socio-functions. For instance, in the music sector, Digital Mark and Digital Francis provided musical e-services by creating interactive and personalized conversations with users and thereby constructing their fan groups to achieve their socio-functions (Whittaker et al., 2025).

Hence, recent scholars have summarized the types of DH, determined the role of DH in the e-service loop and thereby intend to clarify DHs' business value and position (Meng, Wu,

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et al., 2025; Y. Wang et al., 2025; Whittaker et al., 2025). The functions of DHs are based on their information-related nature. With respect to content creation, DHs attempt to deliver information to their customers, acting as assistants, virtual influencers or brand representors in e-service, forming an ecosystem on a digital platform (Meng, Bie, et al., 2025; Thomas et al., 2025; Zhang & Zhu, 2025). For example, B. Wang et al. (2025) used experiments to reveal that virtual influencers use high-arousal language to improve their purchase intention. Moreover, expectancy violation, emotion, and the cognitive effort of a virtual influencer affect DHs' recommendation effects when they are shopping (Nissen et al., 2025). Hence, DH has several effects (persuasion effect, communication effect, endorsement effect, etc.) through interactions with customers in e-service (Meng, Wu, et al., 2025). Such a revelation of the DHs effect in e-services raises the question of whether DHs are effective in e-services (Zhou et al., 2024). That is, the underlying viewpoints of these studies indicated that customers would not entirely accept DHs while they were shopping.

The lack of acceptance of DHs hinders their application in e-service and thereby slows the pace of DH advancement and embedding in real society. This is contrary to the nature of STS. Thus, how to address this issue becomes significant and necessary for current managerial scholars. According to STS, understanding how humans and technology can successfully interact and construct relationships to enable the social functions of DHs in e-service is currently a key focus for improving customer acceptance. Hornbæk and Oulasvirta (2017) argue that human–computer interaction is a form of human–machine relationship (HMR) that explores how HMRs are constructed and maintained, thereby offering solutions to the challenge of realizing the social functions of DHs. Current research on HMRs focuses on how to balance the roles of humans and machines to achieve successful interactions, with the establishment of harmonious human–machine relations seen as the foundation for DHs to perform their social functions (Lee, 2024).

In summary, DHs are types of STSs that exhibit certain social functions inherent to humans. To address the acceptance issue of DHs in e-service, the realization of these social functions is built upon the foundation of harmonious HMRs. Therefore, it is essential to explore how to promote the realization of the social functions of DHs from the perspective of constructing HMRs.

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Customer commitment to DH from a CRM perspective

AI agents can enable DHs to exist independently and separate them from natural humans (Xu et al., 2024; Zhu & Yi, 2024). Extensive research on human–machine affect (Q. Chen et al., 2025), human–machine collaboration models (N. Chen et al., 2024), human–machine value alignment (Khamassi et al., 2024), and human–machine feedback loops (Cimini et al., 2020) illustrates the shift in DHs from being completely lifeless entities to individuals with certain agency. This new form of a DH may cause many affective factors (Q. Chen et al., 2025; Lin, 2024).

Related studies on DH have emphasized the significance of the parasocial relationship between customers and DHs in DH applications. Given that parasocial relationships have been widely employed to explain social media influence (Yuan & Lou, 2020), celebrity marketing (Chung & Cho, 2017), and live-streaming user behavior (Lim et al., 2020), applying this theory to explain customer relationships with DHs is inevitable. Rehman et al. (2025) explored the relationships among DHs' credibility, parasocial interaction and intention to purchase products.

In e-service, CRM, represented by parasocial relationship management, lies at the center of improving customer acceptance and sustaining customer satisfaction (Liu & Wang, 2025; Lou et al., 2023). With respect to DH, parasocial relationship construction is functional and affective. Firms utilize DHs' parasocial relationships to not only improve service quality but also manage their enterprise image, further delivering utilitarian value to customers (Xue et al., 2025). Standing by customers, this construction stirs customers to affect firms or brands. Through DH's timely interactions, customers may fall in love with DHs and thereby become attached to the related firms or brands (Elsharnouby et al., 2023; Vikas Arya et al., 2023). In summary, customers accept and be satisfied with a DH in e-service needs its parasocial relationship construction and management, further stimulating the functional and affective value of DH for firms and customers, respectively (Kim et al., 2023; Lee, 2024).

Furthermore, this reflects that the parasocial relationship is a complex combination of both economic and affective elements. However, current studies on parasocial relationships do not consider these features of parasocial relationships (Chung & Cho, 2017; Tsiotsou, 2015). Moreover, commitment–trust theory reflects that commitment is the core of any relationship (Badrinarayanan & Ramachandran, 2024).

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Commitment between DHs and customers in e-services integrates these features and precisely reflects the complex parasocial relationship management process. Commitment is a type of relationship that originally stems from the relationship between employees and organizations (Alvarez et al., 2023; Roy et al., 2022). According to Meyer et al. (2012), commitment shapes individuals' psychological attachment and reflects the extent to which people internalize or accept the beliefs of and feedback from the object. Moreover, whereas trust reflects a short-term relationship (Elbeltagi & Agag, 2016) and engagement reflects behavioral aspects of interactions (O'Brien et al., 2022), commitment reflects a customer's intention to stay in a long-term relationship with their objects, both behaviorally and psychologically. In the context of DHs, commitment reflects long-term parasocial relationships, including behavioral and psychological aspects.

There are no restrictions on the objects of commitment; whether they are natural humans or brands, individuals can form commitment relationships with them. There are three main types of commitment (Reichers, 1985). The first is continuance commitment, which indicates the willingness to build a long-term relationship with the object. The second is affective commitment, which reflects people's affective investment in the object and their willingness to maintain a relatively stable relationship with it. The third is normative commitment, which indicates that people will follow their moral and normative boundaries when they manage their relationships with the object. These three types of commitment explain the relationship between the subject and the object from both economic and affective perspectives.

In this context, on the basis of commitment–trust theory, customer commitment (to a DH) refers to the intention of customers to build long-term and stable relationships with DHs, including continuance, affective, and normative commitments to the DH. In summary, the realization of the social functions of DHs requires examination from the perspective of customer commitment.

Approaching customer commitment under a utilitarian and hedonic framework

According to studies of customer commitment in e-services, its functional and affective features require us to explore its driving mechanism in terms of its two aspects. Specifically, in terms of the functional aspect of customer commitment, customers are willing to maintain a

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parasocial relationship with DHs in e-services because they perceive the value of e-services in DHs. In terms of affect, customers are inclined to form relationships with DHs because these DHs meet customers' psychological needs in e-service. Therefore, the utilitarian and hedonic frameworks provide a theoretical basis for understanding customer commitment in e-services.

Research on the utilitarian and hedonic frameworks dates back to the work of Holbrook and Hirschman (1982). Their experimental study revealed that utilitarian products are designed to provide utilitarian value to customers, whereas hedonic products offer enjoyment and self-fulfillment value. Thus, utilitarian value links functionality with a brand or product and is purely economic, whereas hedonic value provides emotional connection and is affective. On this basis, the utilitarian and hedonic frameworks are closely related to customers' decisions and feelings, influencing both their cognition and affect. For example, Yuan et al. (2022) reported that when users perceive both utilitarian and hedonic values in artificial intelligence, they are more likely to accept AI-enabled shopping convenience.

Current related studies have begun to rely on this framework to explain the effects of intelligence technology. For instance, Li et al. (2025) detected mainly AI ethical decision-making through a utilitarian and hedonic framework. Longoni and Cian (2022) proposed a word-of-machine effect with a utilitarian and hedonic framework and reported that AI recommenders are more competent than human recommenders in the utilitarian realm and less competent than human recommenders in the hedonic realm are. Obviously, scholars pay more attention to the comparison between the utilitarian and hedonic aspects of AI. This perspective offers insights into exploring DHRM.

In the construction of customer commitment in e-service, two key aspects that customers focus on during their interaction with DHs are the external appearance of the DH and the quality of the information (Meng, Wu, et al., 2025). The aesthetic quality of a DH's appearance affects the customer's psychological state. For example, most customers prefer to psychologically engage with well-designed DHs, and they tend to interact more with those whose appearance is aesthetically pleasing. Thus, DH appearance images offer customers hedonic value in terms of aesthetics (Nagy & Koles, 2014; Vikas Arya et al., 2023). The utilitarian value of DHs primarily depends on the informational value that is expressed during their interaction with customers (Ivanov et al., 2023; Luo et al., 2018).

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Hypothesis development

The utilitarian and hedonic values of DHs influence customers' cognition and affect the e-service loop. Customer experience reflects such impacts. Customer experience is a center of service offering and is defined as a state of affective and cognitive response triggered by external stimuli (Waqas et al., 2021). That is, when DHs embed into the e-service loop, customers initially interact with them, and the features of DHs change the environment of the e-service, acting as a stimulus.

E-services are shaped by interactive information (Rowley, 2006). Aligned with DHs' functions and the features of their parasocial relationships, the information they generate and the external appearance they possess are basic interactive information in DH-embedded e-services. Therefore, during the interaction between customers and DHs in e-services, the appearance of the DH (hedonic value) and the usefulness of the information it conveys (utilitarian value) serve as external factors that enhance the formation of the customer experience in e-services (Iman Khalid A-Qader et al., 2016; Waqas et al., 2021). It can be assumed that a DH with a well-designed image and valuable information will greatly enhance the customer experience during interactions. Thus, the hedonic and utilitarian values of DHs have a positive effect on the customer experience, although the paths of influence may differ. Customer experience can be categorized into sensory, affective, behavioral, and cognitive experiences (Brakus et al., 2009). Sensory and affective experiences are customers' affective experiences, which represent an affective state, whereas behavioral and cognitive experiences are customers' economic experiences, which represent a cognitive state. Therefore, the hedonic value offered by DHs is more likely to influence customers' sensory and affective experiences, whereas their utilitarian value more significantly affects customers' behavioral and cognitive experiences. Accordingly, we propose H1 and its sub-hypotheses:

H1. DHs enhance the customer experience through their appearance images and information quality in terms of both hedonic and utilitarian aspects in e-service.

H1a. The value of DH information positively affects customer sensory, affective, behavioral, and cognitive experiences in terms of the utilitarian value of e-services.

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H1b. The DH appearance image positively influences customer sensory, affective, behavioral, and cognitive experiences in terms of hedonic value in e-services.

As noted, customer commitment includes three distinct types of commitment (Patrucco et al., 2024). The first type is continuance commitment, which refers to the customer's willingness to establish a long-term relationship with the DH. The second type is affective commitment, which refers to the customer's willingness to invest emotionally in the DH. The third type is normative commitment, which describes the customer's intention to adhere to social norms and ethical standards while building a relationship with the DH. These commitments represent the conditions for establishing a stable relationship between the customer and the DH, with the DH fostering the development of this commitment through interaction with the customer. First, DHs can provide sensory experiences to customers through their appearance and voice. Second, customers often invest in interactions with DHs, reflected in affective preferences and the fulfillment of affective needs. Finally, customers form a complete understanding of DHs through actual interactive behaviors. Therefore, the overall experience that DHs provide to customers helps promote the establishment of a relationship between the customer and the DH. Accordingly, we propose H2 and its sub-hypotheses:

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H2. Customer experience with DHs positively affects the construction of customer commitment.

H2a. Customers' sensory experience positively influences their continuance, affective, and normative commitment to DHs.

H2b. Customers' affective experience positively influences their continuance, affective, and normative commitment to DHs.

H2c. Customers' behavioral experience positively influences their continuance, affective, and normative commitment to DHs.

H2d. Customers' cognitive experience positively influences their continuance, affective, and normative commitment to DHs.

Previous studies have shown that DHs provide both hedonic and utilitarian value to

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customers, primarily determined by the DH's appearance and the quality of information transmission, thus enhancing the formation of the customer interaction experience. A well-designed DH appearance image and effective information transmission value enhance customer sensory, affective, behavioral, and cognitive experiences. Following this, customers assess whether interactions with DHs can establish a stable value commitment, leading to the formation of a relatively stable relationship. In this process, the information quality and appearance image of the DH provide utilitarian and hedonic value, respectively. As a result, they serve as the starting point for establishing and maintaining the relationship between the DH and the customers, working together to shape customer commitment. Customer experience plays a mediating role in this process of constructing commitment. On this basis, we propose H3 and its sub-hypotheses.

H3. DH appearance image and information quality consistently influence the construction of customer commitment through the customer experience in terms of both utilitarian and hedonic aspects.

H3a. The DH appearance image and information quality of humans consistently influence the construction of customer commitment.

H3b. Customer experience with DHs plays a mediating role in the construction mechanism of customer commitment.

Based on the above hypotheses, we construct the theoretical model for this paper, as shown in Figure 1.

[Figure 1]

Methods

Data collection

This study uses a survey method to validate the theoretical model. Since this study investigates DH relationship management, it is necessary to selectively choose and collect samples for this research. According to e-service research, DHs act as virtual assistants, livestream hosts, and salespeople, among others, in an e-service loop on a platform (Blut, 2016). Hence, we targeted customers who had experienced multiple roles of DHs in e-service on a

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platform. For example, customers in China can enjoy watching and shopping services on Kuaishou; they can interact with @Guan Xiaofang, a virtual host of Kuaishou in the livestream channel. To ensure that the sample was valid, at the beginning of the questionnaires, we asked participants whether they had interacted with DHs on a digital platform to enjoy e-services. If they had no experience with such interactions, they were excluded from the sample.

We used several methods for recruiting participants. For example, using the "snowball" sampling method in related WeChat groups we reach out to customers with such experience. We also utilized Wenjuanxing and Prolific for data collection assistance. We distributed a total of 1,000 questionnaires and used a time-lagged method, asking participants to fill out the questionnaire at different intervals. The questionnaire was divided into three waves, with a two-week interval between each wave. In the first wave, the participants were asked to complete questions related to the digital human appearance image and digital human information quality; in the second wave, they filled out an instrument on customer experience; and in the third wave, they completed one on customer commitment. Each time that a questionnaire was completed, we conducted quality checks, including attention checks, reverse questions, and time duration control (>180 seconds). Ultimately, 703 valid responses were collected, yielding a valid response rate of 70.3%. In addition, the data collection strongly avoided common method bias (CMB). In this study, we also employed Harman's single-factor test. According to Podsakoff et al. (2003), these data were not polluted by CMB ($Var\%_{component\ 1} = 30.851, Cumulative\%_{component\ 1} = 30.851$).

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Measurements

The scales used in this study are all derived from scales published in high-quality academic journals. We measured the DH appearance image (Keller, 1993) for 6 items ($\frac{CMIN}{DF} = 2.263;\ RMR = 0.016;\ GFI = 0.991;\ TLI = 0.986;\ CFI = 0.991;\ RMSEA = 0.042;\ Cronbach's\ \alpha = 0.83$). DH information quality was measured using 2 items (Yoo & Donthu, 2001) ($Cronbach's\ \alpha = 0.679$). Customer experience was measured by 9 items, with 3 items for sensory, 2 items for affective, 2 items for behavior, and 2 items for intellectual ($\frac{CMIN}{DF} = 2.257;\ RMR = 0.028;\ GFI = 0.985;\ TLI = 0.971;\ CFI = 0.983;\ RMSEA = 0.042;\ Cronbach's\ \alpha = 0.76$) (Brakus et al., 2009). Customer commitment was measured with 11 items, with 3 for affection, 5 for normative, and 3 for continuance ($\frac{CMIN}{DF} = 3.287;\ RMR = 0.045;\ GFI = 0.968;\ TLI = 0.956;\ CFI =$

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0.97; $RMSEA = 0.057$; $Cronbach's\ \alpha = 0.864$) (Allen & Meyer, 1990). Table 1 shows the specific details and related data of the questionnaire. As shown in Table 1, the sample follows a normal distribution (absolute skewness < 3; absolute kurtosis < 10), thereby meeting the assumptions required for subsequent regression analysis.

[Table 1]

Control variables

In this study, we controlled demographic properties on the basis of the aforementioned sample composition. The control variable assignments are as follows: for gender assignment, 2 = sexual minority, 1 = male, and 0 = female. For age assignment: 1 = under 18; 2 = 18--25; 3 = 26--30; 4 = 31--40; 5 = 41--50; and 6 = over 50. For education level assignment: 1 = high school or below; 2 = high school or associate degree; 3 = bachelor's degree; 4 = student; and 5 = other occupations. For monthly income assignment, 1 = less than 1,000 CNY; 2 = 1,000--3,000 CNY; 3 = 3,000--5,000 CNY; 4 = 5,000--8,000 CNY; and 5 = more than 8,000 CNY.

Samples

Table 2 presents the demographic characteristics of the sample in this study. In Table 2, we can observe that the sample is appropriate for our study. For example, the gender of the sample represents the actual follower base of the digital human, with females constituting the majority (N=411, Rate=58.5%). Additionally, youth make up the majority of the sample (N_{18-30} =363, Rate=51.5%), representing the actual followers. Overall, the sample is appropriate.

[Table 2]

Descriptive analysis

This study first conducted descriptive analysis and discriminant validity testing on the sample. Table 3 displays the correlation coefficients and discriminant validity between the variables. All the variables have an AVE greater than 0.5, and a CR value greater than 0.7 indicates that the variables exhibit good discriminant validity. A significant positive correlation is found between DH appearance image, DH information quality, customer experience, and customer commitment, making it appropriate for further statistical analysis.

[Table 3]

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Before hypothesis testing, we examined the multicollinearity among the variables. By calculating the variance inflation factor (VIF), we find that there is no multicollinearity among the variables, meeting the prerequisites for regression analysis ($VIF_{max} = 2.078 < 10$). Furthermore, we use structural equation modeling to examine the fitness of the entire model. Indicators have shown that the entire model is appropriate ($CMIN/DF = 2.395$; $RMR = 0.054$; $GFI = 0.914$; $TLI = 0.907$; $CFI = 0.92$; $RMSEA = 0.045$).

Results

The main effect

In this section, several statistical methods were adopted to verify the hypotheses. Conventional structural equation modeling (SEM) takes advantage of cross-sectional data rather than panel data (Vaithilingam et al., 2024). Because time-lagged data were collected, SEM is not suited for statistical analysis. The MEM is among the main statistics for addressing panel data (Y. Chen et al., 2022; Shi et al., 2023). Thus, fixed effects are reported for examining the hypotheses. The MEM results are organized in Tables 4 and 5.

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[Table 4]

Table 4 illustrates the relationships between DH appearance image, DH information quality, and customer experience. Both DH appearance image (in terms of hedonic value) and DH information quality (in terms of utilitarian value) can enhance customer experience. However, the enhancement mechanism is complex. Compared with the DH appearance image, the DH information quality primarily enhances the customer intellectual experience ($\beta = 0.157, p < 0.05$) without affecting the affective experience ($\beta = 0.072, p > 0.1$). Thus, H1a is partially supported. Compared with the DH information quality, the DH appearance image has a greater effect on enhancing the customer experience. However, since DH appearance does not affect intellectual experience ($\beta = 0.144, p > 0.1$), H1b is also partly supported. Thus, H1 is partly supported. To examine H2 and its sub-hypotheses, Table 5 is shown as follows.

[Table 5]

According to the results in Table 5, the different experiences provided by DHs with varying value orientations have distinct effects on customer commitment. Continuance

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commitment is unaffected by any customer experience, whereas affective commitment is influenced solely by the customer affective experience ($\beta = 0.381, p < 0.01$). Normative commitment is marginally influenced by both customer affective and customer behavioral experiences ($p < 0.1$). Therefore, H2a and H2d are not supported, whereas H2b and H2c are partially supported, and H2 is partially supported.

Congruence effects brought by the utilitarian and hedonic frameworks

Because the utilitarian and hedonic frameworks have been utilized for constructing theoretical models, a further question we are curious about is the congruence effects brought about by both aspects. RSM is generally utilized to detect the congruence effect within variables; hence, the effects of the utilitarian and hedonic framework can be outlined by RSM rather than fixed effects or other statistics (Edwards & Cable, 2009; Schönbrodt et al., 2018). Overall, to compare the utilitarian and hedonic aspects of customer commitment construction and detect the congruence effect of these two aspects, RSM can be utilized to test H3 and its sub-hypotheses. Additionally, bootstrap mediation analysis was also adopted for analyzing the mediation effect. The results of the RSM are shown in Figure 2.

[Figure 2]

[Table 6]

Table 6 shows the results of the surface tests in RSM. The results indicate that both the hedonic and the utilitarian values of DHs can consistently increase customer commitment. Specifically, along the consistency line, the outer points in Figure 3 (DHAI-low, DHIV-low) are lower than the inner points (DHAI-high, DHIV-high). However, for the right-side points (DHAI-high, DHIV-low), the position is higher than that of the left-side points (DHAI-low, DHIV-high), indicating that the enhancement efficiency of the hedonic value of DHs in improving customer commitment is greater than that of their utilitarian value. Thus, H3a is supported. Moreover, we used bootstrapping to test the mediating effect of customer experience. Table 7 displays the results.

We found that the 95% confidence intervals for the direct mediation effects do not include zero ($CI_{95\%-MHI} = [0.1042,\ 0.2099]$; $CI_{95\%-MHV} = [0.1061,\ 0.1817]$), thus supporting H3b.

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[Table 7]

Discussion

With respect to e-services, current studies have explored the DHRM and have attempted to address this issue. Given that DHs are functional and affective and deliver utilitarian and hedonic value to customers when they are embedded into an e-service loop, how to approach a more effective DH is crucial for DHs' business application. In terms of DHs' socio-features, this study is based on STS and explores the construction of customer commitment in terms of both utilitarian and hedonic aspects. Although several unsupported hypotheses exist, several contributions and practical implications are offered.

Discussion of unsupported hypotheses: failures in DH relationship management

Although several hypotheses are unsupported, we can conclude three points of achieving DHs' acceptance in e-service on the basis of our findings. First, when customers enjoy e-services, a significant way to build relationships with DHs is to affect them. According to the MEM and RSM results, it is only on one path that the DH appearance image and information quality increase the customer affective experience, resulting in customer affective commitment, can achieve customer commitment construction in the e-service. This path illustrates that, in e-service offerings, customers intend to build spiritual relationships with DHs, thereby confirming that DHs could be supportive affectively (Q. Chen et al., 2025; Lin, 2024).

Second, unsupported hypotheses potentially reveal features of relationship construction between customers and DHs in e-services. According to the above literature review, a successful relationship reflects customer acceptance of DHs. However, the unsupported hypotheses address a failure in long-term relationship construction on DHs. According to our findings, continuous commitment is not driven by customer experience in e-service, and customer commitment is seldom influenced by customer experience in long-term relationships in e-service, which indicates that the real relationship between DHs and customers is short term in most e-service contexts.

To explain why DHs failed to construct or maintain long-term relationships with customers, the findings further validate that people are not accustomed to regarding DHs as companions. Given that the DH appearance image has a greater impact on customer experiences (with the correlations of the model being greater than those of the DH information quality), customers' initial intent in building a relationship with DHs is driven by their well-designed appearance image. That is, individuals do not truly regard DHs as human beings. Rather, their interest is driven primarily by the appealing external design. This suggests that customers' perceptions of DHs remain superficial in nature. These findings further clarify the aforementioned observation

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that customers are reluctant to form continuance commitments with DHs; rather, they are willing only to engage in long-term relationships with DHs at an affective level.

In association with the results of RSM, DHs are more hedonic than utilitarian. In association with the affective construction, the findings of this study explain how the relationship between DHs and customers is comprehensively constructed in terms of utilitarian and hedonic value. These findings further reveal that during the process of use, customers may initially be attracted to the human-like attributes of DHs, leading to a desire to build deeper emotional and psychological connections.

However, there are instances in which this emotional relationship may exceed its intended limits in e-service. Given that customer affective experience and customer behavioral experience marginally positively influence customer normative commitment, when customers interact with DHs in their e-service journey, they intend to standardize this interaction. According to the definition of customer normative commitment, DHs are supposed to be ethically aligned with our society in e-service. However, the marginal significance indicated that such alignment is not as important as we used to think; several people still attempted to build nonstandardized relationships with DHs in e-service.

These analyses of DHs' commitment construction present challenges for DHs' application in e-service. This study reveals that the lack of customer attention to the functional utility of DHs may lead to a deviation in their application within e-service contexts. After all, DHs retain artificial artifacts, and their long-term value relies on functional capabilities. However, given the current reality that not everyone is able to appropriately interact with machines (Lee, 2024), how to effectively leverage the utilitarian role of DHs in e-service requires thoughtful consideration from DH developers. Furthermore, if the above issue cannot be resolved, it will be difficult to enhance the customer acceptance of DHs through long-term relationship management, ultimately hindering the integration of DHs into e-services.

In summary, at present, customers' perceptions of DHs are superficial, and they do not place sufficient emphasis on the utilitarian performance of DHs within e-services. This directly results in customers' unwillingness to establish continuance commitment with DHs, leading to problems such as low embedding of DHs in e-services and the failure to build long-term relationships. In the context of short-term relationships, customers are willing to engage with DHs hedonically in e-service, thereby developing an affective relationship that may somewhat "cross boundaries". Therefore, at present, when DHs function as STSs, their primary need is hedonic value rather than utilitarian value.

Theoretical contributions

This study offers several theoretical contributions. Initially, this study contributes to DH theoretical studies. Recent studies have begun to solve DHs' acceptance problems by observing

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DHs' failure in several specific service contexts (Franke & Groeppel-Klein, 2024; Whittaker et al., 2025). Furthermore, how to promote the realization of DH social functions is currently a key focus in DH research (Bae et al., 2024; Seymour et al., 2024). Nevertheless, these studies not only neglected the relationship (parasocial) perspective to address the DH relationship management issue but also failed to embed DH into e-services. To address such theoretical gaps, this study initially proposes a theoretical model of customer commitment from the perspective of utilitarian and hedonic aspects, exploring DHs' underlying mechanisms. By introducing commitment theory, this study clarifies that the interaction between DHs and customers is not merely a one-way tool usage relationship, but a multidimensional commitment relationship built on both utilitarian and hedonic values.

Second, the findings of this study contribute to research on HMRs in e-services. Prior studies of e-services have underscored long-term HMRs' impact on them (Blut, 2016); in particular, in the context of HMR research, the main body of past theoretical input has demonstrated that trust, rather than commitment, determines most HMRs (Kim et al., 2023; Spence, 2019). However, trust is a relatively short-term concept for measuring the extent of risk that a customer takes while using machines (Li & Bitterly, 2024; Seymour et al., 2024). In addition, engagement has attracted the attention of HMR scholars, who emphasize the behavioral rather than the psychological aspect of HMR (Bai et al., 2024; O'Brien et al., 2022).

With respect to the studies of the trust and engagement of HMRs, two research flaws are reflected. On the one hand, prior studies lacked a long-term perspective for observing how customers constructed relationships with DHs; on the other hand, the term "HMR" is complex and beyond behavior, but current studies do not reflect this characteristic (Jamye K. Foster et al., 2022; Youn & Jin, 2021; Zhou et al., 2024). Hence, this study introduced commitment, collected longitudinal data, and utilized the MEM to examine DH relationship management in e-services. Specifically, we conducted an in-depth analysis of the characteristics and usage scenarios of DHs, explored the issue of real relationship construction on the basis of their hedonic and utilitarian values, and further analyzed the distinct roles these values play in building customer relationships, thereby expanding the application of this framework.

Finally, this study contributes to customer commitment and experience. In terms of customer commitment in the context of DH, anthropomorphism (Mark Coeckelbergh, 2022) and emotional attachment (Q. Chen et al., 2025) were regarded as crucial antecedents of DH commitment. However, they failed to attach customer experience to customer commitment and neglected the role of customer experience in e-service (Romero et al., 2025; Waqas et al., 2021). Additionally, the role of different types of customer experience requires further exploration in the context of DH research. In this study, varied customer experience serves not only as an antecedent of customer commitment but also as a bridge for building commitment between DHs and customers. Through the mediation effect test in this study, we found that the hedonic

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and utilitarian values of DHs contribute to the mixed construction of customer commitment through the various experiences that they offer to customers. DHs use the customer experience as a mediator to combine both hedonic and utilitarian values in building customer commitment. This systematically reveals the layers of interaction and the pathways to relationship realization between natural humans and DHs, deepens the psychological and emotional mechanisms in HMRs and provides the latest empirical evidence for human–machine communication research.

Practical implications

This study provides guidance for the design, development, and application of DH technology in customer service. From a design perspective, our research shows that to establish a stable commitment relationship with customers, DHs must simultaneously satisfy both utilitarian and hedonic values. This means that DHs should not only demonstrate efficiency and customer-friendliness in providing solutions and navigating service processes but also cultivate an emotional connection with customers. Designers should therefore aim to balance instrumental functionalities—such as clarity of communication and responsiveness—with emotional elements such as tone, facial expressions, and conversational warmth, thereby delivering personalized and human-like service interactions. From a development perspective, this study highlights the contextual variability in the perceived value of DHs. In customer service, emphasis should be placed on enhancing utilitarian value—such as accuracy in problem resolution, professional demeanor, and process efficiency—while sufficient hedonic value should be integrated to foster approachability and customer comfort. Developers are encouraged to tailor the functionalities, appearance, and interaction styles of DHs on the basis of specific service tasks and varying customer expectations to optimize communication effectiveness and service outcomes. From an application perspective, this study reveals that customers' affective and behavioral experiences are critical to the construction of digital human–customer commitment. Accordingly, when digital human technologies are implemented in customer service, organizations should prioritize the continuous refinement of interaction processes on the basis of data analytics and customer feedback. This includes dynamically adjusting the digital human's linguistic expressions, emotional responsiveness, and behavioral cues to enhance emotional resonance and behavioral alignment with customers. Ultimately, such efforts can strengthen customer engagement, increase customer satisfaction, and foster long-term customer loyalty toward the service brand.

Conclusions, limitations and future research

Conclusions

Our findings indicate that for DHs to integrate more smoothly into e-services with customers, they must simultaneously provide value to customers in terms of both hedonic and

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utilitarian aspects. This relationship construction approach is hybrid. The results of the MEM and RSM indicate that during the process of constructing DH–customer commitments, the impact of the hedonic value of DHs is greater, and the efficiency of relationship building increases faster. Moreover, the role of the utilitarian value of DHs is largely consistent with that of hedonic value. Additionally, the data suggest that while DHs are successful at generating short-term affective commitment, they fail to generate long-term loyalty. This implies that customers currently view DHs as "flings" or temporary tools rather than long-term partners. These findings not only deepen the understanding of the role of utilitarian and hedonic frameworks in technology acceptance and usage but also reveal the complexity and multidimensionality of the customer interaction relationship, driving the innovative applications of this framework.

Limitations and future research

This study has the following limitations. First, although the DH appearance image selected in this study is well known in China, its representativeness may be limited, as attitudes and usage patterns toward DHs may vary across different cultural backgrounds and social groups. Second, this study did not comprehensively collect general characteristics of customers' interactions with DHs, such as the duration of usage. Finally, although this study compares the roles of the utilitarian and hedonic values of DHs in the construction of DH–customer commitments, the DH itself, as a technical system, may present more complex layers and dimensions in its appearance image and functionality. Future research could adopt a cross-cultural comparative perspective, selecting DHs from multiple cultural backgrounds to increase the representativeness of the sample, considering individual difference factors, and further enriching the model of this study. Additionally, this study failed to control the multiple roles played by DH in the theoretical model. Thus, future studies should focus not only on the multiple roles of DHs but also on how different types of DH appearance images influence various contexts.

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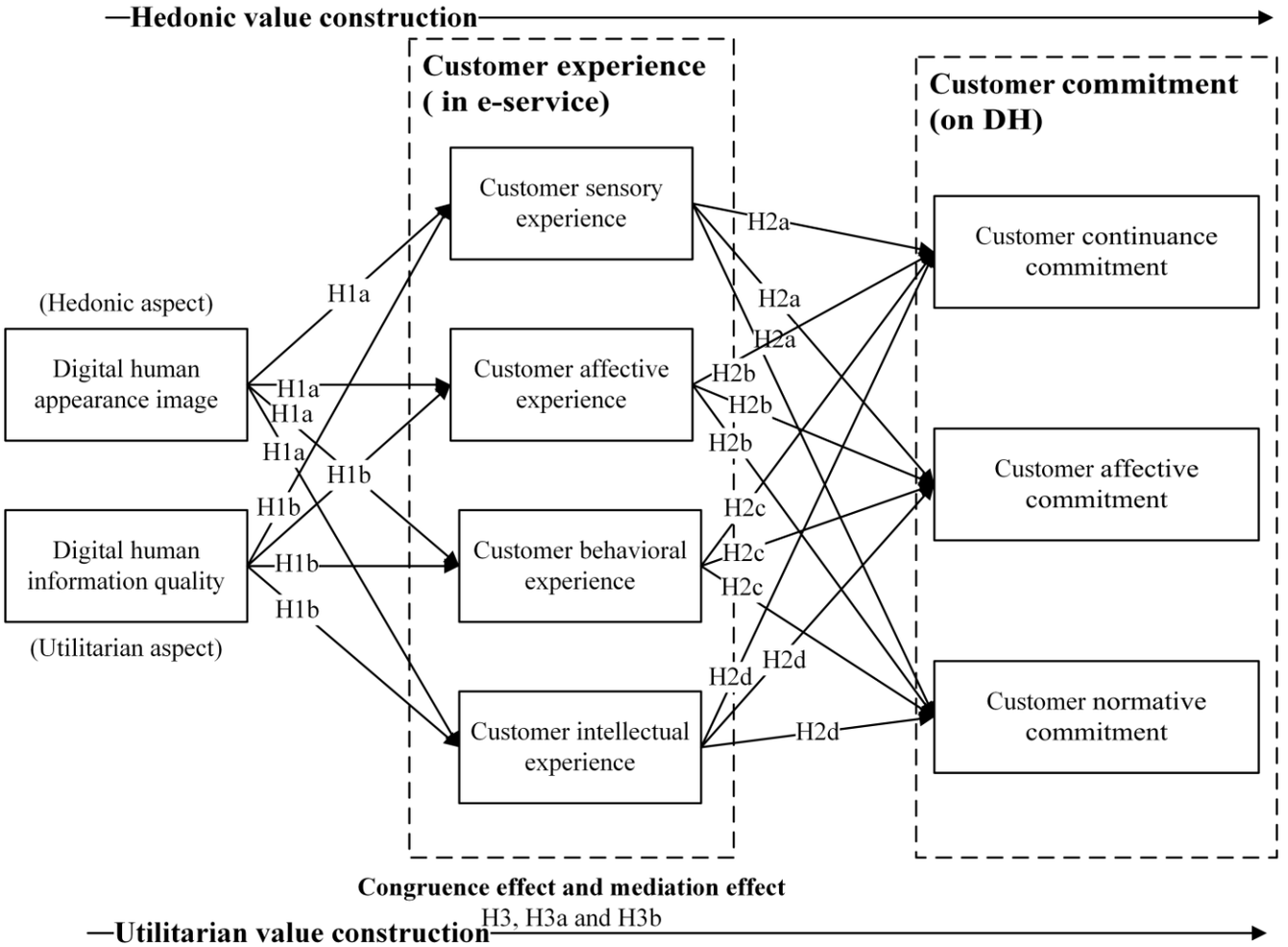
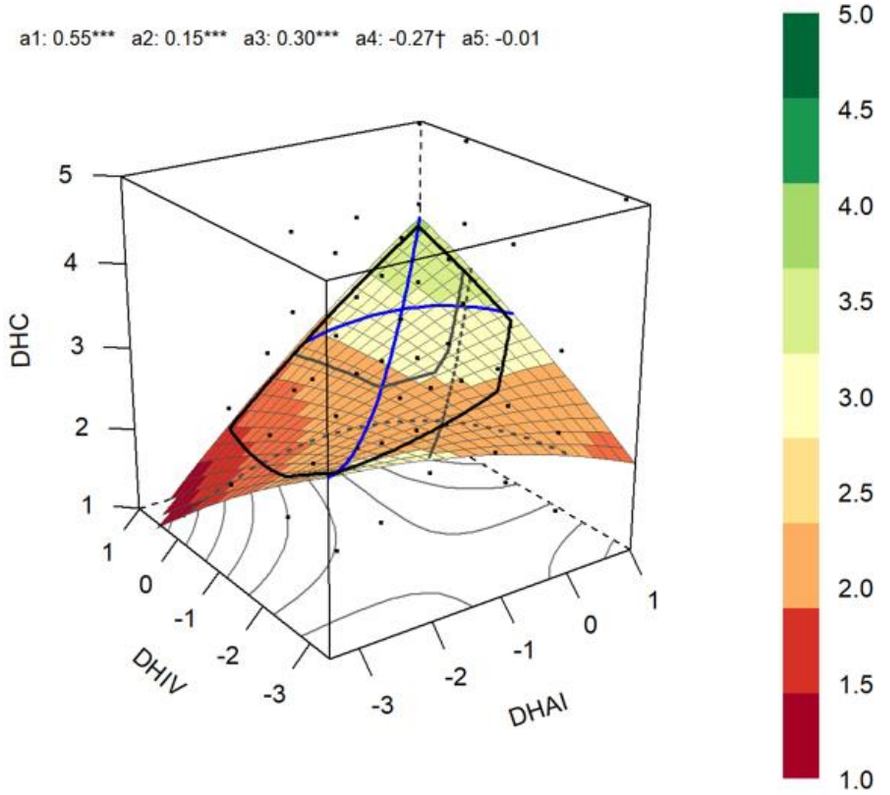


Figure 1 Theoretical framework of digital human-user commitment construction



***Note:** N=703; DHC represents customer commitment (to DHs); DHAI represents digital human appearance image; DHIV represents digital human information quality

Figure 2 Congruent analysis by RSM

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Table 1. Items and descriptive statistics of variables.

Variables	Items	Mean	SD	Skewness	Kurtosis
Digital human appearance image	I think this digital human appearance to be unique.	3.95	0.93	-0.71	0.151
	I think this digital human appearance has a very distinctive personality.	4.04	0.87	-0.703	0.189
	I can quickly recall some features of this digital human appearance.	3.97	0.913	-0.777	0.48
	I can quickly recall this digital human's appearance.	4.26	0.813	-1.027	0.895
Digital human information quality	I believe the intangible value of this digital human appearance gives me reasons to interact with it.	3.78	0.965	-0.484	-0.249
	I find this digital human appearance very appealing and likable.	3.93	0.891	-0.493	-0.288
	During the process of enjoying the e-service, the information quality conveyed by this digital human is particularly good.	3.93	0.874	-0.572	0.015
	During the process of enjoying the e-service, the information functionality conveyed by this digital human is particularly good.	3.93	0.848	-0.507	-0.03
Customer Experience (Sensory)	This digital human leaves a strong visual or other sensory impression on me in my e-service journey.	4.17	0.814	-0.804	0.433
	I find this digital human very interesting from a sensory perspective in my e-service journey.	3.97	0.907	-0.685	0.169
	This digital human is very appealing to my senses in my e-service journey.	4.07	0.903	-0.785	0.199

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Customer Experience (Affective)	Emotionally, I have a very strong feeling toward this digital human in my e-service journey.	4.02	0.885	-0.651	0.055
	This digital human is able to evoke people's emotions in my e-service journey.	3.84	0.988	-0.545	-0.223
Customer Experience (Behavioral)	Interacting with this digital human allows me to engage in real activities and be mindful of my behavior in my e-service journey.	3.93	0.938	-0.697	0.095
	This digital human focuses on the user's practical action experience in my e-service journey.	3.9	0.917	-0.64	0.127
Customer Experience (Intellectual)	I have given much consideration to whether to interact with this digital human in my e-service journey.	3.51	1.131	-0.479	-0.448
	Interacting with this digital human often makes me think.	3.27	1.085	-0.147	-0.59
Customer commitment (Affective)	I consider the issues of this digital human as my own issues.	3.04	1.173	0.106	-0.806
	I feel like I am friend of this digital human.	3.22	1.148	-0.17	-0.727
	I have an emotional attachment to this digital human.	3.49	1.061	-0.234	-0.661
Customer commitment (Normative)	I want to interact with other digital human, but there are too few suitable options.	2.94	1.173	0.116	-0.793
	The disadvantage of stopping the interaction of this digital human now is that I cannot find suitable substitutes.	3.25	1.13	-0.132	-0.751
	I choose to interact with this digital human because it provides me with the greatest benefits.	2.96	1.145	0.035	-0.789

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Customer commitment (Continuance)	I feel uncomfortable interacting with other digital humans.	3.36	1.1 38	-0.284	-0.668
	Loyalty is important, and I will continue to interact with this digital human.	3.53	1.0 63	-0.429	-0.43
	Even if offered some extra benefits, I would not attached to other digital human.	2.82	1.1 67	0.191	-0.797
	Maintaining loyalty to this digital human aligns with my values.	2.85	1.2 24	0.085	-0.975
	Sticking with this digital human is better than frequently switching to other digital humans.	2.99	1.1 51	-0.034	-0.789

***Note:** N=703; SD represents standard deviation.

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Table 2. Psychometric properties of the sample.

Content	Categories	N	%
Gender	Male	258	36.7
	Female	411	58.5
	Non binary gender	34	4.8
Age	Below 18	58	8.3
	18-25	239	34
	26-30	124	17.6
	31-40	117	16.6
	41-50	136	19.3
	Above 50	29	4.1
Education	Less than high school (Junior college)	111	15.8
	High school (Junior college)	229	32.6
	Bachelor	268	38.1
	Post-graduates	95	13.5
Occupation	Government agency or public institution	37	5.3
	Enterprise employee	207	29.4
	Individual industrial and commercial households	95	13.5
	Student	210	29.9
	Other	154	21.9
Salary (monthly)	Under USD 136 (CNY 1,000)	163	23.2
	USD 136 (CNY 1,000) – USD 410 (CNY 3,000)	125	17.8
	USD 410 (CNY 3,0000) – USD 684 (CNY 5,000)	114	16.2
	USD 684 (CNY 5,000) – USD 1095 (CNY 8,000)	118	16.8
	Above USD 1095 (CNY 8,000)	183	26
Total		703	100

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Table 3. Descriptive analysis and discriminate validity of variables.

Variables	Mean	S.D	AVE	CR	(1)	(2)	(3)	(4)
Digital human appearance image (1)	3.99	0.66	0.55	0.88	(0.742)			
Digital human information quality (2)	3.93	0.75	0.76	0.86	0.520**	(0.872)		
Customer experience (3)	3.83	0.56	0.64	0.84	0.594**	0.456**	(0.800)	
Customer commitment (4)	3.07	0.66	0.65	0.85	0.452**	0.312**	0.452**	(0.806)

*Note: N=703, **p<0.005 (two-tailed); S.D represents standard deviation; AVE and CR provide evidence of discriminate validity of the model in this study; The diagonal value is the square root of A.V.E.

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Table 4. MEM results for hypotheses examination (fixed effect on customer experience)

Variables	Customer sensory experience (Model 1)		Customer affective experience (Model 2)		Customer behavioral experience (Model 3)		Customer intellectual experience (Model 4)	
	Estimates	SE	Estimates	SE	Estimates	SE	Estimates	SE
Intercepts	1.436***	0.183	0.798***	0.205	1.607***	0.272	2.178***	0.307
Independent variables								
Digital human appearance image	0.513***	0.04	0.629***	0.045	0.363***	0.058	0.144	0.066
Digital human information quality	0.15***	0.035	0.072	0.039	0.228***	0.051	0.157*	0.058
Control variables								
Gender	-0.028	0.04	0.023	0.045	0.035	0.059	-0.002	0.068
Age	-0.008	0.022	0.023	0.024	0.093**	0.032	0.037	0.038
Education	0.005	0.025	0.003	0.028	-0.01	0.037	-0.005	0.043
Occupation	-0.003	0.019	0.045	0.021	-0.029	0.029	0.039	0.034
Salary (Monthly)	0.013	0.021	0.031	0.023	-0.079	0.031	-0.07	0.036
ICC (1)	0.633		0.591		0.843		0.964	

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Adjusted ICC (1)	0.943	0.88	1	1
R^2	0.329	0.328	0.157	0.036
N	703	703	703	703

***Note:** SE represents standard error; ****p<0.001*, ***p<0.01*; **p<0.05*

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Table 5. MEM results for hypotheses examination (fixed effect on customer commitment)

Variables	Customer continuance commitment (Model 1)		Customer affective commitment (Model 2)		Customer normative commitment (Model 3)	
	Estimates	SE	Estimates	SE	Estimates	SE
Intercepts	1.453***	0.277	0.226	0.25	0.373	0.243
Independent variables						
Customer sensory experience	-0.016	0.155	0.1	0.142	0.165	0.136
Customer affective experience	0.065	0.155	0.381**	0.141	0.243†	0.136
Customer behavioral experience	0.215	0.155	0.213	0.141	0.229†	0.136
Customer intellectual experience	0.119	0.155	0.115	0.141	-0.004	0.136
Controls						
Gender	0.013	0.155	0.006	0.141	0.077	0.136
Age	0.014	0.155	0.056	0.14	0.151	0.136
Education	-0.071	0.155	-0.116	0.141	-0.116	0.136
Occupation	-0.005	0.155	0.018	0.141	0.024	0.136
Salary (Monthly)	0.045	0.155	-0.019	0.14	0.001	0.136
ICC (1)	0.762		0.691		0.707	
Adjusted ICC (1)	0.779		0.763		0.779	
R^2	0.022		0.095		0.092	
N	703		703		703	

***Note:** SE represents standard error; ****p<0.001*; ***p<0.01*; †*p<0.1*

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Table 6. Response surface examination results

Indicators of surface test	Estimates	SE	P	Significance
a_1	0.548	0.049	0.000	***
a_2	0.154	0.044	0.000	***
a_3	0.302	0.076	0.000	***
a_4	-0.272	0.156	0.080	+

*Note: N=703; SE represents standard error; ****p<0.001*; *+p<0.1*; NS represents none significance

Table 7 Mediation analysis by Bootstrapping

Paths	Mediation	effect	SE	LLCI	ULCI
DHAI→CE→CC	Direct	0.2603	0.0397	0.1825	0.3382
	Indirect	0.1568	0.0273	0.104	0.2114
DHIQ→CE→CC	Direct	0.0995	0.032	0.0366	0.1642
	Indirect	0.1424	0.0192	0.1065	0.1816

*Note: N=703; SE represents standard error; LLCI represents lower level of 95% confidence interval; ULCI represents upper level of 95% confidence interval.

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets generated during and analysed during the current study are available from the corresponding author on reasonable request.

Ethical approval

All research procedures were approved by the Ethics Committee of Dali University (Approval No. HSSECDLU-20240501) on May 1, 2024. The research was conducted in line with the 1964 Declaration of Helsinki.

Informed consent

Consent was provided through the cover letter attached to the questionnaire. All participants were fully understanding the informed consent form. This process was conducted between Jun and July 2024. All authors confirm that informed consent was obtained from every participant involved in the study. Participants voluntarily agreed to take part in the research, fully informed that their anonymity would be ensured, that their data would be used solely for research purposes, and that there were no risks associated with their participation. Data were recorded and numbered in Arabic numerals, which did not include participants' names.

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Author contributions

Lingzhi Fang: conceptualization, validation, writing—original draft preparation, project administration.

Tan Cheng: methodology, validation, software.

Heng Yang: methodology, writing—review and editing, supervision, funding acquisition, writing—review and editing.